

CH 101

Chemistry

(3-1-0-8)

Structure and Bonding; Origin of quantum theory, postulates of quantum mechanics; Schrodinger wave equation: operators and observables, superposition theorem and expectation values, solutions for particle in a box, harmonic oscillator, rigid rotator, hydrogen atom; Valence Bond and Molecular Orbital Theories; Hydrogen Molecule; Hybridization; Molecular Symmetry; Electronic Spectroscopy and Lasers. Chemical Thermodynamics and Chemical Kinetics.

Coordination compounds: ligand, stereochemistry, crystal field and molecular orbital theories; Bioinorganic chemistry and organometallic chemistry; Chemistry of materials.

Stereochemistry of more than two stereo-centers, R&S and E&Z nomenclature, Conformation of cyclohexane and 1,2-disubstituted cyclohexane; Pericyclic reactions; Bioorganic chemistry: proteins, enzymes, carbohydrates, nucleic acids and lipids; Natural products: classification and origin of terpenoids, alkaloids and steroids. Macromolecules (polymers); Solid phase synthesis; Green chemical processes. Modern spectroscopic techniques in structural elucidation of organic compounds (UV-vis, IR, NMR).

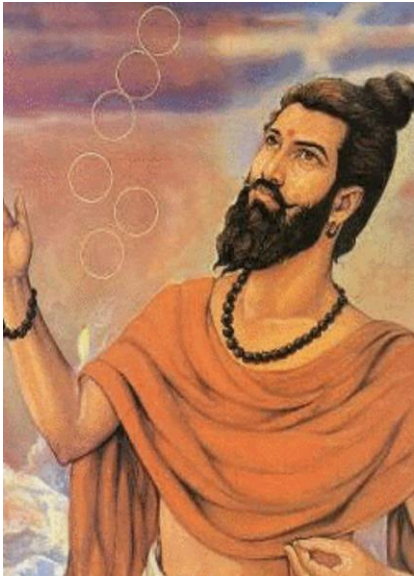
Text Books

1. P. W. Atkins, *Physical Chemistry*, 5th Ed., ELBS, 1994.
2. C. N. Banwell, and E. M. McCash, *Fundamentals of Molecular Spectroscopy*, 4th Ed., Tata McGraw-Hill, 1962.
3. **F. A. Cotton, and G. Wilkinson, *Advanced Inorganic Chemistry*, 6rd Ed., Wiley Eastern Ltd., New Delhi, 1972, reprint in 1988.**
4. **D. J. Shriver, P. W. Atkins, and C. H. Langford, *Inorganic Chemistry*, 2nd Ed., ELBS ,1994.**
5. S. H. Pine, *Organic Chemistry*, McGraw-Hill, 5th Ed., 1987

References:

1. I. A. Levine, *Physical Chemistry*, 4th Ed., McGraw-Hill, 1995.
2. I. A. Levine, *Quantum Chemistry*, EE Ed., prentice Hall, 1994.
3. G. M. Barrow, *Introduction to Molecular Spectroscopy*, International Edition, McGraw-Hill, 1962
4. **J. E. Huheey, E. A. Keiter and R. L. Keiter, *Inorganic Chemistry: Principle, structure and reactivity*, 4th Ed., Harper Collins, 1993**
5. L. G. Wade (Jr.), *Organic Chemistry*, Prentice Hall, 1987.

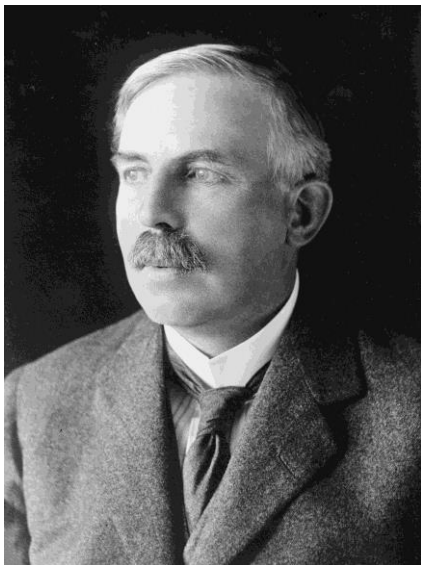
An Atom and its Constituents



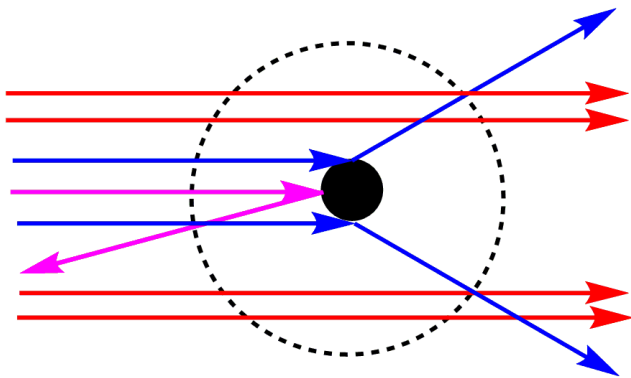
500 BC



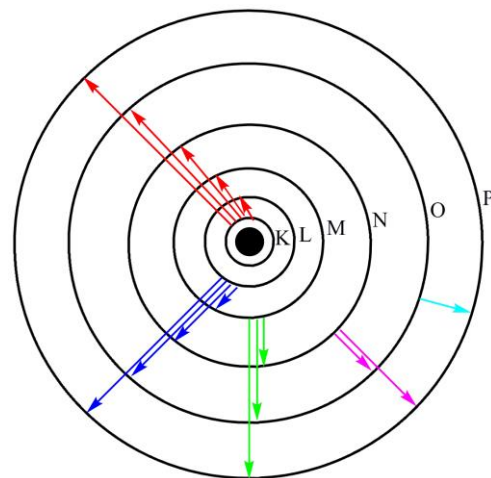
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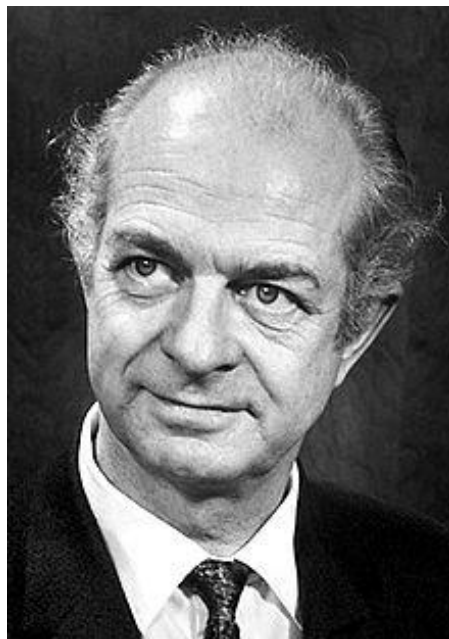
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Orbit *versus* Orbital

Orbit	Orbital
Orbit is well-defined circular path.	Orbital represents the space where the probability of finding an electron/electrons is maximum.
It represents two dimensional motion of an electron.	It represents three dimensional motion of an electron.
Denoted by K, L, M, N,...	Denoted by s , p (p_x , p_y , p_z), d (five d orbitals),
The maximum no. of electrons in an orbit is $2n^2$.	The maximum no. of electrons in an orbital is 2.
Orbit is circular in shape because it comes from Bohr model	Orbitals have different shapes (except s orbital). de Broglie, Heinesberg, Sommerfield,.. That is way it determines the geometry of a complex.
It represents particle nature.	It represents wave nature.

Quantum numbers

Bohr-Sommerfeld atomic model and its later extensions, four quantum numbers (n , l , m , s) were introduced to explain the spectral features of atoms.

(A) The principle quantum number, n .

This quantum number determines the energy level of an atom. The energy expressed as a function of n ,

$$E_n = -\frac{Z^2 m e^4}{8 \epsilon_0^2 n^2 h^2}$$

According to the above equation energy should decrease with increase of n value. Due to negative sign the energy also increases with increase in the value of n .

(B) The Azimuthal or angular momentum quantum number, l :

The Azimuthal or angular momentum quantum number determines the orbital angular momentum of an electron through the following relation. This specifies the shape of an atomic orbital, its bonding and bond angles.

$$L^2 = l(l + 1)h^2$$

l may have any integral values between 0 to $(n-1)$. For $l = 0$, defines s orbital. $l = 1, 2, 3$, etc defines, p, d, f , etc respectively.

(C) The magnetic quantum number, m_l :

This quantum number determines the orientation of the angular momentum vector of an electron in the presence of external magnetic field along a specific axis (let consider along z axis).

$$L_z = m_l h$$

For a given l value m_l will be in between $+l$ to $-l$.

$$\text{i.e. } m_l = 2l + 1$$

Example: for p orbital, $l = 1$, therefore, $m_l = +1, 0, -1$. This indicates p orbital will have three different directions orientations in the presence of external magnetic field.

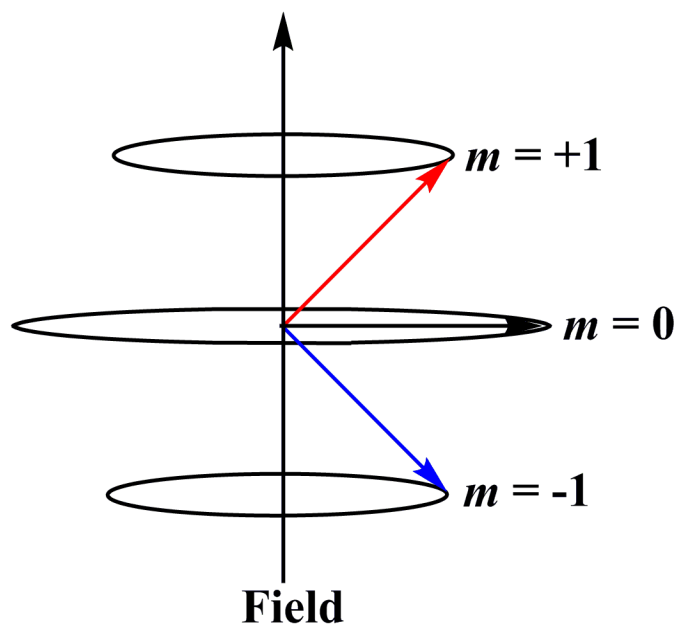


Figure . The orientation of orbital angular momentum vector in the presence of external magnetic field for $l = 1$.

(D) Spin projection quantum number, m_s :

An electron spins about its own axis. Because of this spinning an angular momentum is generated that is completely different from the orbital angular momentum.

This angular momentum is expressed by spin projection quantum number (m_s).

$$L_z = m_s h$$

For a given s value m_s will have value in the range of $+s$ to $-s$.

An electron has $s = \frac{1}{2}$ spin, Hence, $m_s = +\frac{1}{2}, -\frac{1}{2}$

Some other examples, $s = \frac{3}{2}$, $m_s = +\frac{3}{2}, +\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}$

$$s = 3, m_s = +3, +2, +1, 0, -1, -2, -3$$

This indicates, $m_s = 2s + 1$

n	l $0 \leq l \leq n - 1$	m_l $-l \leq m_l \leq l$	Orbital type	Total no. of orbitals $2l + 1$	Maximum no of electrons $2(2l + 1)$
1	0	0	s	1	2
2	0	0	s	1	2
	1	+1, 0, -1	p	3	6
3	0	0	s	1	2
	1	+1, 0, -1	p	3	6
	2	+2, +1, 0, -1, -2	d	5	10
4	0	0	s	1	2
	1	+1, 0, -1	p	3	6
	2	+2, +1, 0, -1, -2	d	5	10
	3	+3, +2, +1, 0, -1, -2, -3	f	7	14

Sequence of energy level

Rules:

(A) Orbital energies increases as $(n + l)$ increases.

(B) If there are two orbitals with the same value of $(n + l)$, the one with the smaller n has the lower energy.

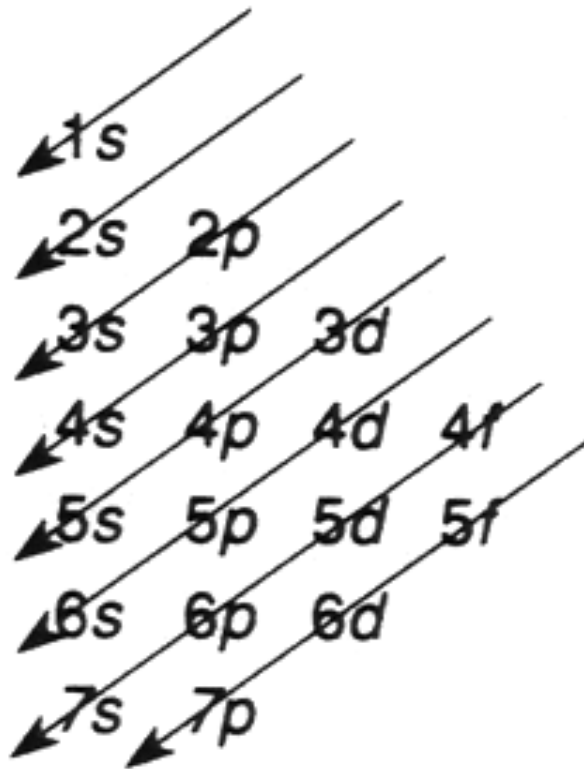


Figure . Energy levels sequence